

SGA 6 – Exposé XIII

Original source scan pages 654–669

5. Finiteness theorems of Néron–Severi type

Theorem 5.1. Let X be proper over a noetherian scheme S . Then the Néron–Severi groups

$$\mathrm{NS}(X_K) = \mathrm{Pic}_{X_K/K}(K) / \mathrm{Pic}_{X_K/K}^\tau(K)$$

of the geometric fibres X_K of X/S are finitely generated. Their ranks and the orders of their torsion subgroups remain bounded.

By noetherian induction it is enough to prove the assertion after replacing S by an arbitrary non-empty open subscheme of S_{red} . We may therefore suppose that $\mathrm{Pic}_{X/S}^\tau$ is represented by an S -scheme of finite type. Then the order of $\mathrm{Tors}(\mathrm{NS}(X_K))$ is the number of connected components of $(\mathrm{Pic}_{X/S}^\tau)_K$, and this number is constant over a non-empty open subset of S .

It remains to bound the rank of $\mathrm{NS}(X_K)$. As in the proof of 4.7 (iii), one reduces to the case where S is integral and X is projective and integral over S . Let r be the common dimension of the fibres. For $r = 0$, respectively $r = 1$, the rank is plainly 0, respectively 1. If $r \geq 2$, one first reduces to the case $r = 2$ and X smooth over S : after shrinking S , choose a linear section Y of X , integral over S and with fibres of dimension 2, and replace X by Y using 4.3. Then, after a flat covering of a non-empty open subset of S , the same 4.3 together with 3.10 and resolution of singularities for surfaces permits replacing X by a smooth S -scheme.

Now, by 4.6 (i) $\Rightarrow$ (iii) and by the finiteness and specialization theorem for ℓ -adic cohomology groups (SGA 4 XVI 2.2), the theorem follows from the following proposition.

Proposition 5.2. Let k be algebraically closed and let X be proper and smooth over k . Then the group $C^p(X)$ of algebraic cycles of codimension p , modulo numerical equivalence, is free of rank at most the Betti number b_{2p} .

Ignoring the action of roots of unity, *étale* cohomology

$$H_{\mathrm{et}}^i(X, \mathbb{Q}_\ell) = \left(\varprojlim_n H_{\mathrm{et}}^i(X, \mathbb{Z}/\ell^n \mathbb{Z}) \right) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \quad \ell \neq \mathrm{char}(k),$$

is a Weil cohomology: it satisfies Poincaré duality and the Künneth formula, and receives a non-zero functorial homomorphism from the ring of algebraic cycles modulo rational equivalence,

$$C^p(X) \longrightarrow H_{\mathrm{et}}^{2p}(X, \mathbb{Q}_\ell(p)).$$

Let $a_1, \dots, a_m \in C^{r-p}(X)$, with $r = \dim X$, be classes whose images form a basis of the $\mathbb{Q}_\ell$ -vector subspace of

$$H_{\mathrm{et}}^{2(r-p)}(X, \mathbb{Q}_\ell(r-p))$$

spanned by algebraic cycles. Consider

$$\alpha : C^p(X) \longrightarrow \mathbb{Q}_\ell^m, \quad \alpha(b) = (\langle b \cdot a_1 \rangle, \dots, \langle b \cdot a_m \rangle).$$

Then α is injective on numerical classes, and the bound on the rank follows.

Remark 5.3. If k is algebraically closed and X is a proper smooth surface over k , then

$$b_2(X) = c_2(X) - 2 + 2b_1(X),$$

where $b_1(X) = 2 \dim(\text{Pic}_{X/k}^0)$ and $c_2(X)$ is the second Chern class, equivalently the Euler–Poincaré characteristic, or the self-intersection number of the diagonal in $X \times X$. Thus $b_2(X)$ has an interpretation independent of cohomology. In characteristic > 0 , the inequality

$$\text{rank } C^1(X) \leq b_2(X)$$

in this form is due to Igusa, before *étale* cohomology, using deep results on the structure of the tame fundamental group of an algebraic curve minus finitely many points.

Theorem 5.4. Let k be algebraically closed and let X be proper over k . Then there exist finitely many closed integral curves $Y_1, \dots, Y_p$ in X such that a subset A of the scheme $\text{Pic}_{X/k}^\tau$ is quasi-compact if and only if the intersection numbers

$$\langle L, Y_i \rangle, \quad L \in A,$$

remain bounded. Moreover p may be taken to be the rank of the Néron–Severi group

$$\text{Pic}_{X/k}(k) / \text{Pic}_{X/k}^0(k).$$

Since $\text{Pic}_{X/k}^0$ is an open quasi-compact subgroup of $\text{Pic}_{X/k}^\tau$, a subset A is quasi-compact if and only if the corresponding set of points in

$$(\text{Pic}_{X/k}^\tau(k) / \text{Pic}_{X/k}^0(k)) \otimes_{\mathbb{Z}} \mathbb{Q}$$

is finite. But τ -equivalence is the same as numerical equivalence, by 4.6, so one may choose curves in X whose numerical classes form a basis of the dual vector space.

6. Appendix: study of (b) -sheaves on $P = \mathbf{P}_k^N$

Let k be algebraically closed and let $P = \mathbf{P}_k^N$. Let $\mathcal{O}_P(1)$ be the canonical invertible sheaf on P and put

$$S = k[x_0, \dots, x_N].$$

6.1. Let F be a coherent sheaf on P , with Hilbert polynomial

$$\chi(F(n)) = \sum_{i=0}^r a_i \binom{n+i}{i},$$

and let $(c) = (c_0, \dots, c_r)$ be a sequence of integers, chosen at least as large as the previously fixed sequence (a) . Put

$$a_{m,i} = \sum_{j=0}^{r-i} a_{j+i} \binom{m-1+j}{j}, \quad c_{m,i} = \sum_{j=0}^{r-i} c_{j+i} \binom{m-1+j}{j}.$$

Finally, for every integer $q \geq 1$, denote by N_q the largest coherent subsheaf of F whose support has dimension $\leq q-1$, and put $F_q = F/N_q$.

Lemma 6.2. Under the conditions of 6.1, consider an exact sequence

$$0 \longrightarrow I \longrightarrow \mathcal{O}_P^{\oplus M} \longrightarrow F(m) \longrightarrow 0$$

with $m \geq 0$. Then I is p -regular, where $p = P_r(c_{m,0}, \dots, c_{m,r})$ and P_r is the r -th (b) -polynomial.

Indeed, $\mathcal{O}_P$ is clearly a $(0, \dots, 0, 1)$ -sheaf, with $N+1$ entries; hence I is a $(0, \dots, 0, M)$ -sheaf by 1.6. Thanks to 2.10,

$$\chi(I(n)) = M \binom{n+N}{N} - \sum_{i=0}^r a_{m,i} \binom{n+i}{i}.$$

The assertion follows from 1.11 and 1.10.

Proposition 6.3. Under the conditions of 6.1, suppose that F is a (b) -sheaf. Then one can calculate, by means of universal polynomials in the a_i , the b_i and N , integers m, M, m_1, M_1 such that there is an exact sequence

$$\mathcal{O}_P(-m_1)^{\oplus M_1} \longrightarrow \mathcal{O}_P(-m)^{\oplus M} \longrightarrow F \longrightarrow 0.$$

Moreover N enters only in the calculation of M_1 .

One chooses m large enough to make $F(m)$ globally generated and to control $h^0(F(m))$ by the polynomials already obtained. If I is the kernel of $\mathcal{O}_P^{\oplus M} \rightarrow F(m)$, Lemma 6.2 supplies a universal regularity bound for I ; taking m_1 beyond that bound makes $I(m_1)$ generated by its global sections, and gives the asserted presentation after twisting back.

Theorem 6.4. If F is a quotient of $\mathcal{O}_P(-m)^{\oplus M}$, with $m \geq 0$, then there are explicit universal bounds $b_0, \dots, b_r$ such that F is a $(b_0, \dots, b_r)$ -sheaf and its successive Hilbert-polynomial coefficients and zero-th cohomology dimensions are bounded by universal polynomials in m, M and N .

The proof proceeds by induction on the dimension of the support and by restricting to a hyperplane. If $H \subset P$ is a hyperplane not containing any associated component, one uses the exact diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I(-1) & \longrightarrow & I & \longrightarrow & I_H & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{O}_P(-1)^{\oplus M} & \longrightarrow & \mathcal{O}_P^{\oplus M} & \longrightarrow & \mathcal{O}_H^{\oplus M} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & F(m-1) & \longrightarrow & F(m) & \longrightarrow & G(m) & \longrightarrow & 0. \end{array}$$

The induction hypothesis applied to the hyperplane quotient, together with Lemma 6.2 for the kernel, gives the stated universal bounds.

Lemma 6.5. Suppose that F has no associated prime cycle of dimension 0.

- i) If $h^0(F) > 1$, then $h^0(F(-1)) < h^0(F) - 1$.
- ii) $H^0(F(-n)) = 0$ for $n \geq h^0(F)$.
- iii) If $0 \rightarrow F(-1) \rightarrow F \rightarrow G \rightarrow 0$ is exact and $H^0(G(-n_0)) = 0$, then $H^0(F(-n_0)) = 0$ and $h^0(F) \leq n_0 h^0(G)$.

The proof begins with a non-zero section of F , compares the subsheaf it generates with its hyperplane restriction, and uses

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F'(-1) & \longrightarrow & F' & \longrightarrow & G' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & F(-1) & \longrightarrow & F & \longrightarrow & G & \longrightarrow & 0. \end{array}$$

Proposition 6.6. If F is a $(b_0, \dots, b_r)$ -sheaf, then the quotients $F_q = F/N_q$ obtained by removing the pieces of support of dimension $< q$ are again (b') -sheaves for explicitly computable

sequences (b') . The proof uses the hyperplane-section diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & N_q(-1) & \longrightarrow & N_q & \longrightarrow & G' & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(-1) & \longrightarrow & F & \longrightarrow & G & \longrightarrow & 0 \\
& & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F_q(-1) & \longrightarrow & F_q & \longrightarrow & G'' & \longrightarrow & 0.
\end{array}$$

For $q = 1$, the support of N_q is zero-dimensional; for $q > 1$, the assertion follows by induction on q and by Lemma 6.5.

Theorem 6.7. The preceding estimates give universal polynomial bounds for the Hilbert polynomials of the successive quotients F_q and for the regularity of the pieces occurring in the filtration by dimension of supports. Thus a bounded numerical datum for F determines a bounded presentation, a bounded regularity, and bounded Hilbert polynomials for all F_q .

Corollary 6.8 (Grothendieck). Let X be projective over a noetherian scheme S , let $\mathcal{O}_X(1)$ be very ample for X/S , and let $\mathcal{F}$ be a bounded family of classes of coherent sheaves on the fibres of X/S . Then there is an integer m such that, for every class $\xi \in \mathcal{F}$ and every coherent sheaf F representing ξ , the sheaf F is m -regular on the corresponding fibre.

After replacing the base field by an algebraic closure of a purely transcendental extension, one may suppose the ground field infinite. The assertion follows from the previous regularity estimates, from 6.10 and 1.8 (i), and from 6.2 applied to $F(n)$ for $n \geq m$.

Definition 6.9. A family of coherent sheaves on projective spaces is said to be bounded if it is contained in a family obtained from finitely many presentations with bounded twists and bounded numbers of generators. Equivalently, after fixing an embedding, the family is represented by a quasi-compact part of the corresponding Hilbert or Quot scheme.

Lemma 6.10. Let F be coherent on $\mathbf{P}_k^N$, with Hilbert polynomial bounded in the preceding sense. If the numerical data (a) and (b) are fixed, then the possible sheaves F satisfying the (b) -conditions form a bounded family. The proof uses Proposition 6.3 to obtain a presentation by free sheaves with bounded twists and bounded ranks, and hence a point of a bounded Quot scheme.

Corollary 6.11. In the situation of 6.10, the dimensions $h^i(F(n))$, the regularity bounds, and the Hilbert polynomials of the subsheaves N_q and quotients F_q are all bounded by universal functions depending only on the fixed numerical datum.

Remark 6.12. The preceding estimates are intentionally effective: they are phrased in terms of explicit universal polynomials, because later applications require not merely boundedness but uniform control over all algebraically closed ground fields and all projective spaces of the prescribed dimension.

Corollary 6.13. Let F vary in a bounded family on projective schemes endowed with a very ample sheaf. Then, after a single integer twist independent of the member of the family, all members are generated by global sections and have vanishing higher cohomology in the Castelnuovo–Mumford range.

Corollary 6.14 (L. Hermann). For every $N \geq 0$ there is a polynomial $R_N(x)$ such that, for every algebraically closed field k , every projective k -scheme X with very ample sheaf $\mathcal{O}_X(1)$, and every coherent sheaf F on X satisfying $\chi(F(n)) = N$ for all n , one has

$$h^0(F) \leq R_N(\deg X).$$

7. Appendix: the Hodge index theorem

We recover and complement here some earlier results by a direct argument.

Theorem 7.1. Let X be an irreducible surface, proper over an algebraically closed field. Suppose that there exists an invertible sheaf H on X such that

$$\langle c_1(H)^2 \rangle > 0$$

(for example H ample). Then an invertible sheaf L on X satisfying

$$\langle c_1(L), c_1(H) \rangle = 0, \quad \langle c_1(L)^2 \rangle \geq 0$$

is numerically equivalent to zero. Consequently, by 4.6, the intersection form on

$$\text{NS}(X) \otimes \mathbb{Q} = (\text{Pic}(X) / \text{Pic}^0(X)) \otimes \mathbb{Q}$$

is non-degenerate of type $(1, \rho - 1)$.

First suppose that X is reduced, hence integral, that H is ample, and that

$$\langle c_1(L), c_1(H) \rangle = 0, \quad \langle c_1(L)^2 \rangle > 0.$$

Arguing by contradiction, choose $m > 0$ such that $H_1 = L \otimes H^{\otimes m}$ is ample, and choose n so that, with

$$N = L^{\otimes n} \otimes H^{-1},$$

one has

$$\langle c_1(N), c_1(H_1) \rangle = n \langle c_1(L)^2 \rangle - m \langle c_1(H)^2 \rangle > 0.$$

At the same time

$$\langle c_1(N), c_1(H) \rangle = -\langle c_1(H)^2 \rangle < 0,$$

and

$$\langle c_1(N)^2 \rangle = n^2 \langle c_1(L)^2 \rangle + \langle c_1(H)^2 \rangle > 0,$$

contradicting the next lemma.

Lemma 7.1.2. Let X be an integral surface, projective over an algebraically closed field. For an invertible sheaf N on X with $\langle c_1(N)^2 \rangle > 0$, the following conditions are equivalent:

- i) for every ample sheaf H , one has $\langle c_1(N), c_1(H) \rangle > 0$;
- (i') there exists an ample sheaf H with $\langle c_1(N), c_1(H) \rangle > 0$;
- ii) there exists an integer $n > 0$ such that $H^0(X, N^{\otimes n}) \neq 0$.

If (ii) holds, an effective divisor D exists with $N \simeq \mathcal{O}_X(D)$, and $\langle c_1(N)^2 \rangle > 0$ forces $D > 0$, giving (i). Conversely, to see that (i') implies (ii), assume $H = \mathcal{O}_X(1)$ very ample. There is a closed integral locally principal curve Y with $H = \mathcal{O}_X(Y)$. Riemann's theorem gives vanishing of the relevant cohomology for $N^{\otimes n}$ for n large. The exact sequence

$$H^0(N^{\otimes n}(m)) \rightarrow H^0(N^{\otimes n}(m)|_Y) \rightarrow H^1(N^{\otimes n}(m-1)) \rightarrow H^1(N^{\otimes n}(m)) \rightarrow 0$$

then implies $H^1(N^{\otimes n}) = 0$ for all sufficiently large n . Hence

$$h^0(N^{\otimes n}) = \chi(N^{\otimes n}) = \frac{1}{2} \langle c_1(N)^2 \rangle n^2 + \dots > 0$$

for $n \gg 0$.

Now suppose in 7.1 that there is an invertible sheaf M on X with

$$\langle c_1(L), c_1(M) \rangle \neq 0.$$

Arguing by contradiction, replace X by the underlying reduced scheme of a Chow presentation; this is allowed by the projection formula. Let ξ be ample on X . Choose $m > 0$ such that $H_1 = \xi^{\otimes m}$ is ample and

$$\langle c_1(L), c_1(H_1) \rangle \neq 0.$$

Then choose integers p, q with $q \neq 0$ and set

$$L_1 = L^{\otimes p} \otimes H_1^{\otimes q}$$

so that

$$\langle c_1(L_1), c_1(H) \rangle > 0, \quad \langle c_1(L_1)^2 \rangle > 0.$$

This reduces to the preceding case.

Finally, in the general case, assume again by contradiction that L is not numerically equivalent to zero. Replace X by the normalization $\tilde{X}$ of X_{red} ; then $\tilde{X}$ has only finitely many singular points, and the replacement is justified by the projection formula and 4.5 (ii). There is a closed integral curve Y on $\tilde{X}$ with

$$\langle c_1(L), Y \rangle \neq 0.$$

The curve Y is locally principal outside a finite set Z . Let $f : X' \rightarrow \tilde{X}$ be the blow-up of the ideal I defining the defect. The ideal $\mathcal{M} = I\mathcal{O}_{X'}$ is invertible and defines a subscheme of X' which is the union of two subschemes Y' and Y'' with $f(Y') = Y$ and $f(Y'') \subset Z$. By the projection formula,

$$\langle c_1(f^*L), c_1(\mathcal{M}) \rangle = -\langle c_1(f^*L), Y' \rangle - \langle c_1(f^*L), Y'' \rangle = -\langle c_1(L), Y \rangle \neq 0.$$

Replacing X, L, H by X', f^*L, f^*H brings us back to the preceding case and completes the proof of 7.1.

Remark 7.2. It is not known whether every irreducible proper surface over an algebraically closed field which carries an invertible sheaf H with $\langle c_1(H)^2 \rangle > 0$ must be projective. Nor is it known whether, for every irreducible proper surface, the intersection form on $\text{NS}(X) \otimes \mathbb{Q}$ is non-degenerate. It follows from 7.1 that, when it is degenerate, it must be negative, not indefinite.

Corollary 7.3. Under the hypotheses of 7.1, let M be an invertible sheaf on X not numerically equivalent to zero, and set

$$a = \langle c_1(M), c_1(H) \rangle, \quad h = \langle c_1(H)^2 \rangle.$$

Then

$$\langle c_1(M)^2 \rangle < a^2/h.$$

Indeed, take $L = M^{\otimes h} \otimes H^{-a}$. Since $\langle c_1(L), c_1(H) \rangle = 0$, Theorem 7.1 gives $\langle c_1(L)^2 \rangle < 0$.

Corollary 7.4. Let X be an irreducible projective scheme of dimension $r \geq 2$ over an algebraically closed field, and let H be ample on X . Then:

i) an invertible sheaf L on X is numerically equivalent to zero if and only if

$$\langle c_1(L) c_1(H)^{r-1} \rangle = 0, \quad \langle c_1(L)^2 c_1(H)^{r-2} \rangle \geq 0;$$

ii) for every invertible sheaf M on X not numerically equivalent to zero,

$$\langle c_1(M)^2 c_1(H)^{r-2} \rangle \langle c_1(H)^r \rangle < \langle c_1(M) c_1(H)^{r-1} \rangle^2.$$

Assertion (ii) follows from (i) as in 7.3. To prove (i), let Y be a closed integral curve in X . It is enough to prove $\langle c_1(L), Y \rangle = 0$. Embed X in projective space by $H^{\otimes m}$, $m \gg 0$. If $I = \bigoplus I_\nu$ is the homogeneous ideal of Y , choose ν_0 such that I_ν generates I for $\nu \geq \nu_0$. Hypersurfaces Z of degree ν containing Y have no common point outside Y . By induction on r and Bertini's theorem, choose $r - 2$ such hypersurfaces $Z_1, \dots, Z_{r-2}$ so that

$$X' = Z_1 \cap \dots \cap Z_{r-2} \cap X$$

is irreducible. Since $\langle c_1(L), Y \rangle = \langle c_1(L|_{X'}), Y \rangle$, it remains to prove that $L|_{X'}$ is numerically equivalent to zero on the surface X' , which follows from 7.1 by the projection formula.

Corollary 7.5. Let k be algebraically closed. Let Y be projective over k , with ample sheaf H , and let X be the zero-scheme of a section of H . Suppose that, for every irreducible component Y^* of Y , all irreducible components of

$$X \cap Y^* = V(\omega|_{Y^*})$$

have dimension > 2 . Then

$$\mathrm{NS}(Y) \longrightarrow \mathrm{NS}(X)$$

is injective. It is enough to assume Y integral, and the assertion follows from 7.4 (i) and the projection formula.

Bibliography for Exposé XIII

- [1] S. Abhyankar, Resolution of singularities of arithmetical surfaces, in *Arithmetical Algebraic Geometry*, Purdue University, Harper & Row, New York, 1965.
- [2] D. Mumford, Pathologies III, *Amer. J. Math.* 1967, pp. 94–103.
- [3] A. Grothendieck and J. Dieudonné, EGA, *Éléments de Géométrie Algébrique*, Publ. Math. IHÉS, 1960.
- [4] A. Grothendieck, FGA, *Fondements de la géométrie algébrique*, extracts from the Séminaire Bourbaki, 1957–1962.
- [5] A. Grothendieck et al., SGA, *Séminaire de géométrie algébrique*, 1960–66, IHÉS.
- [6] O. Jussila, Exposé X, Formalism of intersections on proper algebraic schemes, in this seminar.
- [7] M. Raynaud, Exposé XII, A relative representability theorem for the Picard functor, in this seminar.